

Running the First Test

A. Installation and Setup

- a. Run setup.bat (or setup.sh) to install the required dependencies into a local virtual environment.

 setup	07/06/2026 15:42	Windows Batch File	3 KB
---	------------------	--------------------	------

- B. Once installation has completed, run run.bat (or run.sh) to launch the application.

 run	07/06/2026 15:42	Windows Batch File	1 KB
---	------------------	--------------------	------

- C. Optionally, create a dedicated folder in which to store the results generated throughout this tutorial.

 Result	10/06/2026 13:30	File folder	
--	------------------	-------------	--

1. Position Classification

sRNA Pipeline - Integrative RNA-Seq & Co-exp...

< 01 Position Classification 02 Co-expression analysis Differential I

GTF file: Test_annotation.gtf Browse...

GTF feature type: exon

GTF attribute key (ID): exon_id

sRNA ID prefix: srn_

Gene ID prefix: gene

sRNA rename prefix: sRNA

Gene rename prefix: Gene

Overlap threshold (fraction): 0.90 ^ v

UTR window distance (bp): 150 ^ v

Output file (.xlsx/.csv): NA_annotation.xlsx Browse...

Run Position Classification

- Select the GTF annotation file Test_annotation.gtf, located in the 00 Test data folder.
- Specify the desired output file.
- Click Run Position Classification.
- Expected output:**

Found 681 sRNAs and 3013 genes.

Category counts:

location

Intergenic 374

Antisense 219

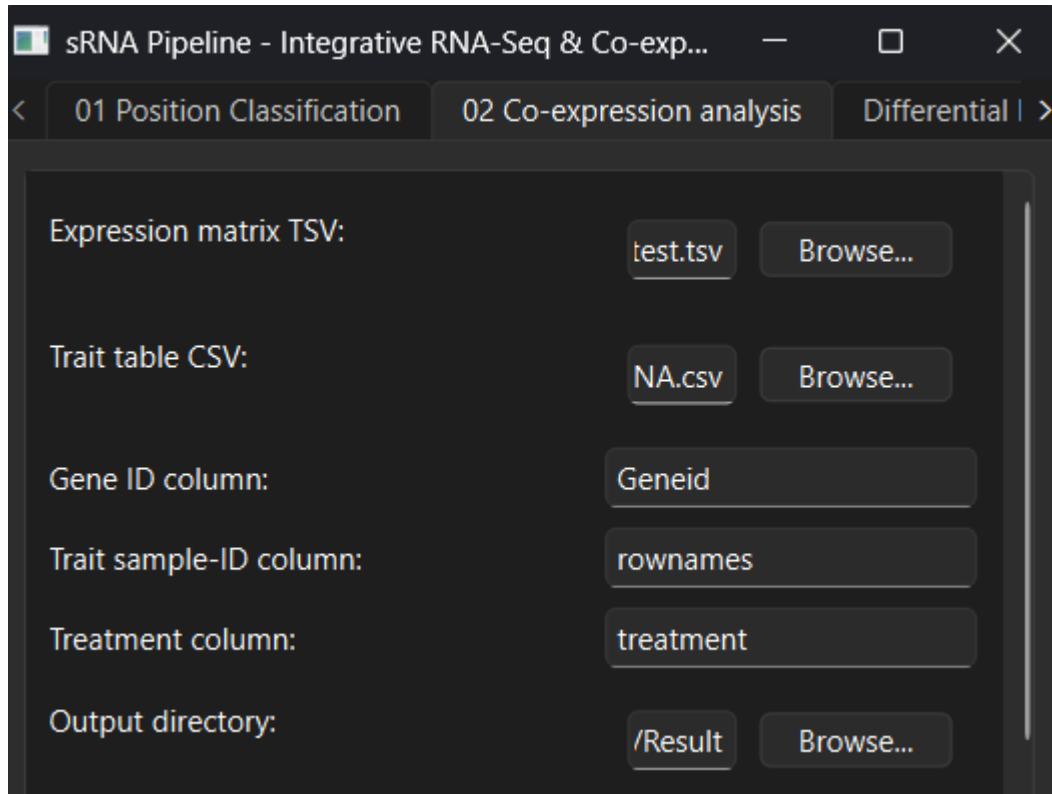
3' UTR 31

5' UTR 30

Intragenic 27

Saved sRNA_annotation.xlsx in the desired output directory.

2. Co-expression Analysis (WGCNA)



sRNA Pipeline - Integrative RNA-Seq & Co-exp...

< 01 Position Classification 02 Co-expression analysis Differential | >

Expression matrix TSV: test.tsv Browse...

Trait table CSV: NA.csv Browse...

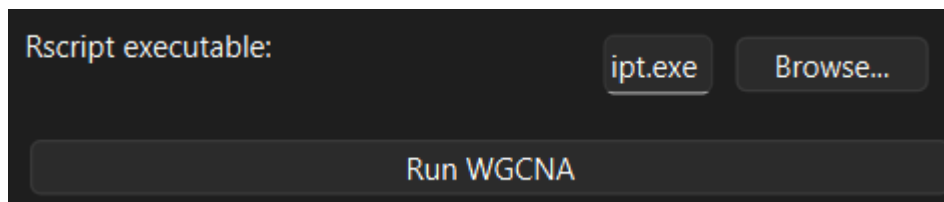
Gene ID column: Geneid

Trait sample-ID column: rownames

Treatment column: treatment

Output directory: /Result Browse...

Input: select Count_test.tsv as the expression matrix (TSV).
Select table_treatment_WGCNA.csv as the trait table (CSV).
Set the output directory.
Adjust the RStudio (Rscript) executable path if required.
Click Run WGCNA.



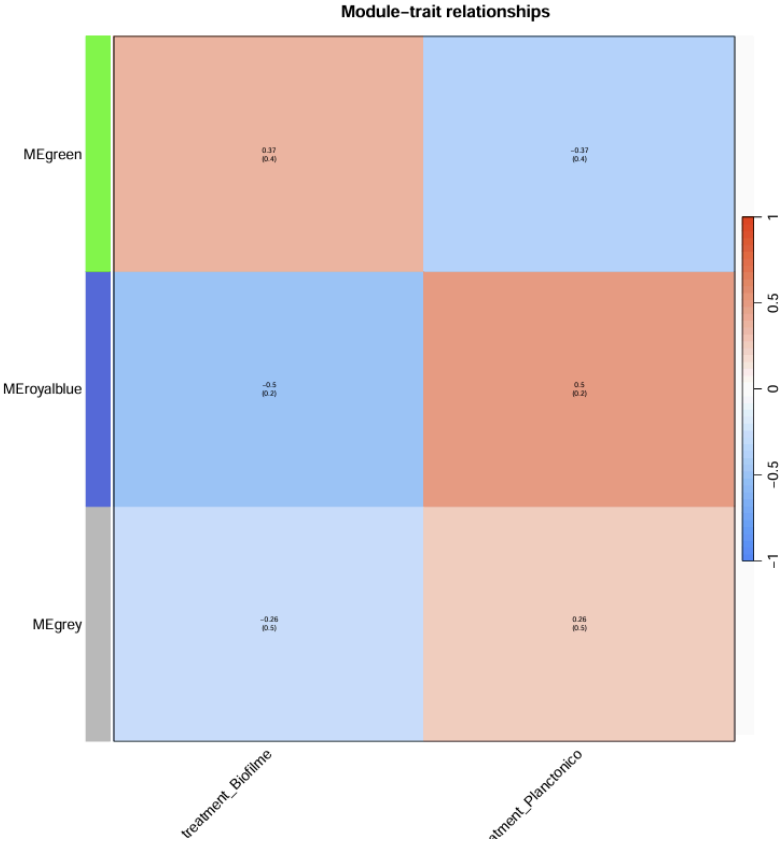
Rscript executable: ipt.exe Browse...

Run WGCNA

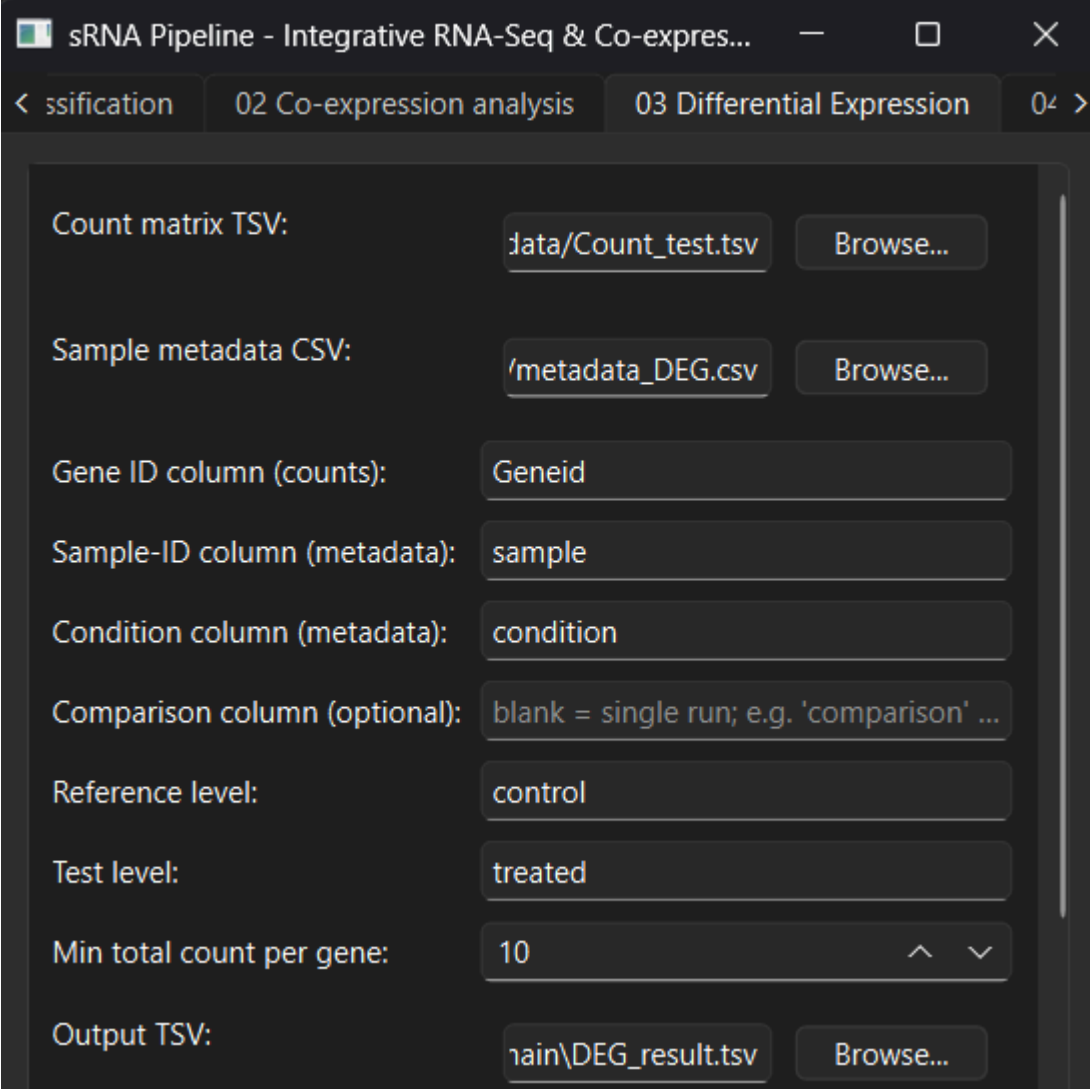
The following files are generated in the results folder:

	top_hub_genes_per_module	10/06/2026 13:30	Comma Separated...	1 KB
	gene_module_membership	10/06/2026 13:30	Comma Separated...	74 KB
	CytoscapeInput-edges	10/06/2026 13:30	Text Document	133,722 KB
	CytoscapeInput-nodes	10/06/2026 13:30	Text Document	71 KB
	gene_dendrogram_with_modules	10/06/2026 13:30	PNG File	153 KB
	Module_Trait_Heatmap	10/06/2026 13:30	Microsoft Edge PD...	6 KB
	soft_thresholding_power_selection	10/06/2026 13:30	PNG File	61 KB
	sample_dendrogram_with_traits	10/06/2026 13:30	PNG File	63 KB
	sample_dendrogram_outlier_check	10/06/2026 13:30	PNG File	54 KB
	WGCNA_configured	10/06/2026 13:30	R Source File	8 KB

A module-trait heatmap is also produced for comparison:



3. Differential Expression (DESeq2)



sRNA Pipeline - Integrative RNA-Seq & Co-expression analysis

< ssification 02 Co-expression analysis 03 Differential Expression 04 >

Count matrix TSV: data/Count_test.tsv Browse...

Sample metadata CSV: /metadata_DEG.csv Browse...

Gene ID column (counts): Geneid

Sample-ID column (metadata): sample

Condition column (metadata): condition

Comparison column (optional): blank = single run; e.g. 'comparison' ...

Reference level: control

Test level: treated

Min total count per gene: 10 ^ v

Output TSV: main\DEG_result.tsv Browse...

Input: set the count matrix TSV to Count_test.tsv.

Set the sample metadata CSV to metadata_DEG.csv.

For this test, set the Comparison column to comparison. This splits the metadata into groups (e.g., A, B, C, D) and runs DESeq2 once per group, producing one DEG file per comparison (DEG_result_A.tsv, DEG_result_B.tsv, ...).

Comparison column (optional): comparison

Set the output directory.

Adjust the RStudio (Rscript) executable path if required.

Click Run DESeq2.

Rscript executable:

bin\x64\Rscript.exe

Browse...

Run DESeq2

Expected output:

Shared dispersion estimated from 8 samples over 3211 genes.
comparison A -> DEG_result_A.tsv [own dispersion] sig padj<0.05: 591
comparison B -> DEG_result_B.tsv [own dispersion] sig padj<0.05: 1804

 DEG_result_A	10/06/2026 13:46	TSV File	367 KB
 DEG_result_B	10/06/2026 13:46	TSV File	373 KB

4. Filtering

The image shows a software window titled "sRNA Pipeline - Integrative RNA-Seq & Co-expression" with four tabs: "01 Position Classification", "02 Co-expression analysis", "03 Differential Expression", and "04 Filters". The "04 Filters" tab is active. It contains a text box with the instruction: "Leave any file blank to skip that step. If the Prediction CSV is blank, sRNA-target pairs are built from the edges file." Below this are four input fields, each with a text box and a "Browse..." button:

- Prediction CSV: `\\A-main\00 Test data\Prediction_test.csv`
- DEG TSV file(s): `\\A-main\DEG_result_A.tsv` (with a "Clear" button next to it)
- WGCNA nodes file: `\\A-main\Result\CytoscapeInput-nodes.txt`
- WGCNA edges file: `\\A-main\Result\CytoscapeInput-edges.txt`

Prediction CSV: select Prediction_test.csv.

Add both DEG_result_A.tsv and DEG_result_B.tsv to the DEG TSV files field.

Add CytoscapeInput-nodes.txt to the Cytoscape Node field.

Add CytoscapeInput-edges.txt to the Cytoscape Edges field.

In the DEG filter configuration, set the minimum strains to 2.

The image shows a "Differential expression (DEG) filters" configuration window. It has two rows of settings, each with a label, a text input field, and up/down arrow buttons:

- padj cutoff: 0.0500
- Min strains consistent: 2

Expected output:

Total predictions: 33996

After energy/probability filter ($\geq 5/5$ of ['E_intaRNA', 'E_Rnaplex', 'E_TargetRNA3', 'Probability_TargetRNA3', 'Probability_sRNARFTarget']): 29662

Consistent DEGs: 572

After DEG filter: 423

After module filter (same): 234

After requiring network edge: 183

After best_per_target selection: 146

Final interactions: 146

The image shows a file explorer bar at the bottom of the application. It contains the following information:

- File icon and name: `filtered_weight`
- Date and time: `10/06/2026 13:52`
- File format: `Comma Separated...`
- File size: `27 KB`